



Sustainable portfolio optimization: A multi-class framework for eco-friendly stocks

Abdul Wahid^a, Oskar Kowalewski^{b,c,*}

^a Department of Accounting and Finance, NUML School of Business, National University of Modern Languages (NUML), H-9, Islamabad, Pakistan

^b IESEG School of Management, Univ. Lille, CNRS, UMR 9221 - LEM - Lille Économie, Management, Lille F-59000, France

^c Institute of Economics, Polish Academy of Sciences, Warsaw, Poland

ARTICLE INFO

JEL Classification in this article:

G11
D40
F21

Keywords:

Portfolio choice
Green stocks
Efficient frontier curve
Dual-class stocks

ABSTRACT

This study investigates the impact of green stocks on portfolio performance over short- and long-term horizons and identifies key determinants influencing these returns. Stocks are categorized as green or non-green based on pre-IPO carbon emissions. The findings reveal that green stocks yield lower returns but exhibit superior portfolio efficiency in the short term compared to non-green stocks. Conversely, in the long run, green stocks outperform non-green stocks, demonstrating greater stability and profitability. Dual-class stocks deliver higher returns, which can be attributed to offshore risks and diversification benefits. Furthermore, parental market return dynamics and local currency exchange rate fluctuations significantly influence the performance of green stocks listed on AIM. Similarly, parental market dynamics and foreign exchange rate variability are critical factors affecting the returns of offshore stocks.

1. Introduction

Sustainable investing has garnered increasing attention from academics, policymakers, and investors. Green stocks, representing companies with low carbon emissions and environmentally responsible practices, have emerged as a significant area of interest, attracting substantial capital inflows due to the growing emphasis on sustainability (Du and Zhang, 2020; Mumtaz and Yoshino, 2021). However, systemic and financial barriers continue to shape investment allocation to green stocks, particularly in production-intensive industries. These barriers include a preference among investors for higher short-term returns and systemic risks linked to the integration of green assets, which diminish their appeal (Horn and Oehler, 2024; Akhtaruzzaman et al., 2023). Additionally, the high upfront costs of adopting sustainable technologies often suppress short-term returns, further limiting green investments' competitiveness compared to traditional alternatives.

Despite the expanding literature on green stock investing, the debate on optimal portfolio allocation remains unresolved. Divergent perspectives persist regarding the trade-off between downside risk mitigation and upside potential. Proponents argue that the stable and predictable returns from green investments enhance portfolio efficiency by reducing volatility and promoting diversification (Aliu et al., 2021; Zhou et al., 2021). Conversely, critics contend that growth-focused investors may prioritize higher-yielding non-green stocks, particularly in bullish markets, even if this involves greater volatility and susceptibility to sharp corrections (Dobrynskaya, 2015; Fisher and D'Alessandro, 2021). This tension underscores the challenge of balancing short-term profitability with long-term sustainability in portfolio construction.

* Correspondence to: IESEG School of Management, 1 Parvis de la Défense, Paris la Défense 92044, France.

E-mail addresses: abwahid.fms@gmail.com (A. Wahid), o.kowalewski@ieseg.fr (O. Kowalewski).

Modern portfolio theory offers a robust framework for addressing this trade-off by emphasizing asset diversification, including green and non-green stocks, to optimize risk and return. Recent advancements underscore the importance of incorporating eco-friendly stocks to mitigate global risks and advance sustainability objectives. While green stocks are associated with lower volatility and reduced risk, their contribution to long-term profitability and portfolio efficiency remains underexplored, especially in markets serving smaller firms and cross-listed equities, such as the Alternative Investment Market (AIM) (Mumtaz and Yoshino, 2021; Akhtaruzzaman et al., 2023).

This study contributes to the growing literature on sustainable portfolio optimization by analyzing the role of green and non-green stocks in enhancing portfolio performance over short- and long-term horizons. Leveraging a novel Greenness Index based on pre-IPO carbon emissions, the study develops a rigorous framework to classify investments and assess their performance across local and dual-class stock categories. The analysis employs data from 312 AIM-listed firms (2006–2020), evaluating the impact of firm-specific, market-specific, and governance factors on returns.

The results reveal that non-green stocks outperform in the short term due to higher volatility and informational asymmetries, while green stocks deliver superior long-term returns, reflecting enhanced stability and alignment with sustainability goals. Dual-class stocks consistently achieve higher returns across both horizons, benefiting from diversification and offshore risk exposure. AIM's regulatory framework, characterized by modest listing requirements and support for small and cross-listed firms, further strengthens the long-term performance of green investments.

The study identifies key determinants of returns. Larger, older firms with established reputations achieve higher buy-and-hold abnormal returns (BHAR) over extended periods, whereas initial underpricing adversely affects long-term performance due to price corrections. Governance factors, including board size, institutional shareholding, and CEO duality, also shape outcomes, with stronger governance positively associated with higher returns.

Our work is most closely aligned with Rahat and Nguyen (2022), who advocate for divestment from non-green stocks to optimize long-term portfolio performance in emerging markets. However, our approach diverges in significant ways. We include both green and non-green stocks, enabling an evaluation of portfolio dynamics across varying horizons. To support this analysis, we introduce a novel Greenness Index that categorizes stocks based on carbon emissions relative to production levels into low green, high green, low non-green, and high non-green classifications, offering deeper insights into investment sensitivity and greenness dynamics.

Our findings illustrate contrasting investor preferences, with low non-green stocks excelling in the short term and green stocks dominating in the long run, emphasizing their stability and alignment with sustainability objectives. This study not only examines allocation strategies across different horizons but also highlights the superior performance of green assets, particularly in shorter timeframes, aligning with evidence from Akhtaruzzaman et al. (2023). Additionally, we observe an increasing preference for short-term investments in non-green stocks, driven by immediate monetary gains over environmental considerations (Horn and Oehler, 2024; Akhtaruzzaman et al., 2023). By addressing key challenges in stock selection and allocation central to modern portfolio theory, our study offers a robust framework for assessing portfolio performance across diverse investment horizons.

The remainder of the paper is structured as follows. Section 2 reviews the literature on portfolio optimization and the performance of green investments. Section 3 describes the data, econometric models, and methodologies employed. Section 4 presents the results and discusses key findings, including the performance of green and non-green stocks within local and dual-class categories. Finally, Section 5 concludes with the study's implications for sustainable investing and directions for future research.

2. Literature Review

The discussion on constructing optimal portfolios has long been a cornerstone of portfolio management literature, with evaluating portfolio performance emerging as a more complex and technical domain that has sparked further debate (Awiszus et al., 2022; Hoque et al., 2020). Modern Portfolio Theory posits that an optimal portfolio can be achieved by diversifying allocations across various asset types, thereby maximizing returns for a given level of risk or minimizing risk for a desired return (Fisher and D'Alessandro, 2021). Portfolio construction facilitates diversification, risk mitigation, and achieving an optimal balance through the risk-return trade-off. The following sections review key aspects of the literature related to sustainable allocation, diversification, risk management, and the dynamics of AIM.

2.1. Sustainable allocation

Portfolio allocation remains a critical component of financial management, enabling investors to achieve desired returns while managing associated risks. Traditional approaches, such as the mean-variance model, have been extensively used for portfolio optimization. However, modern methods incorporating psychological and technological factors have gained prominence. Behavioral finance, for example, explains portfolio allocation through cognitive biases, which may lead to precarious decision-making (Antony, 2020). To counter this, diversified portfolios are recommended to mitigate risks associated with individual biases.

Rahat and Nguyen (2022) provide evidence of the financial viability of sustainable allocation strategies, demonstrating that divestment from high-carbon industries and reallocation to green portfolios yield superior risk-adjusted returns in emerging markets. By incorporating environmental metrics such as carbon intensity and green revenue proportions, their methodology highlights how sustainability-oriented allocation frameworks can outperform traditional approaches. These findings underscore the potential for integrating environmental considerations into portfolio construction, expanding the scope of classical models like mean-variance optimization.

Gunjan and Bhattacharyya (2023) argue for a hybrid approach, blending classical and modern techniques, emphasizing the need

for rational asset allocation combined with innovative methodologies. Advanced models, such as the Fuzzy-based AHP (Analytic Hierarchy Process) and FANP (Fuzzy Analytic Network Process), have been employed to account for financial market volatility and enhance decision-making (Jawad et al., 2024; Rahiminezhad Galankashi et al., 2020). These approaches fall under Multi-Criteria Decision-Making (MCDM) frameworks.

In addition, artificial intelligence (AI) and machine learning have emerged as transformative tools for portfolio allocation. Predictive techniques, such as Long Short-Term Memory (LSTM) networks, have been used to forecast stock prices and proactively guide investment decisions (Chaweewanchon and Chaysiri, 2022; Thakkar and Chaudhari, 2021). These innovations enable investors to address complex variables and make data-driven allocation choices.

2.2. Diversification and sustainable investing

Diversification remains a fundamental strategy to minimize risk and enhance returns. Among sustainable financial instruments, green bonds have been found to offer unique diversification benefits. Akhtaruzzaman et al. (2022) provide evidence that portfolios incorporating green bonds not only lower systemic risk but also outperform non-green counterparts during crises. These findings underscore the evolving role of green instruments in advancing diversification strategies, complementing traditional stock portfolios with sustainable financial products. Moreover, the principles of diversification evident in green bonds are increasingly mirrored in the construction of stock portfolios, where the integration of environmentally aligned assets contributes to risk mitigation and return optimization.

Rahat and Nguyen (2022) further highlight the diversification potential of green portfolios, showing that they consistently outperform black and universal portfolios across risk-adjusted metrics in emerging markets. By incorporating environmental metrics such as carbon intensity and green revenue proportions, their study demonstrates the value of green assets in enhancing portfolio resilience. These findings align with the broader trend of integrating sustainable investments to achieve both financial and environmental objectives, reinforcing the role of green portfolios in modern diversification strategies.

Huang et al. (2024) analyzed portfolio strategies using data from Taiwan-listed companies, concluding that diversification across multiple assets improves performance while mitigating risk. Among the strategies evaluated, the certainty equivalent return (CER) emerged as a superior metric over the Sharpe ratio in volatile markets. Similarly, advanced optimization methods, such as Monte Carlo simulations, have been utilized to predict stock behavior and enhance diversification (Ta et al., 2020).

Islamic finance has also contributed to diversification strategies. Hoque et al. (2020) observed that while conventional and Islamic portfolios perform equally at the market level, Islamic portfolios outperform at the sector level, demonstrating their potential as an alternative diversification avenue.

Cryptocurrencies have added complexity to diversification practices. Aliu et al. (2021) found that cryptocurrencies offer high returns but exhibit significant volatility and limited diversification benefits compared to traditional assets. Similarly, Demiralay and Bayracı (2021) noted a decline in cryptocurrency diversification benefits since 2017 due to increased market integration.

Finally, Zaimovic et al. (2021) emphasized the evolving number of stocks needed for effective diversification, influenced by economic crises such as the Global Financial Crisis and the COVID-19 pandemic. These studies highlight the importance of adaptive diversification strategies while recognizing the enduring relevance of traditional diversification methods.

2.3. Risk and Trade-Off

Effective risk management is critical to achieving portfolio efficiency. Fjesme (2020) underscored the dangers of concentrated investments for inexperienced investors, while experienced investors may find concentrated portfolios rewarding. Advanced predictive techniques, such as LSTM models, significantly enhance risk management and portfolio performance by improving stock price forecasts (Cipiloglu Yildiz and Yildiz, 2022).

The refinement of mean-variance models through statistical advancements has been shown to improve accuracy in risk-return estimates, reducing errors in covariance and mean return calculations (Dai and Kang, 2022). Ethical investments, such as ESG-focused portfolios, present unique trade-offs between performance and specific risks, particularly in emerging markets (Beloskar and Nageswara Rao, 2024). These findings underline the need for dynamic, adaptive risk management strategies that align with changing market conditions.

2.4. Dynamics of AIM and portfolio diversification

The AIM offers unique opportunities for portfolio diversification, especially through offshore listings. AIM has grown into a significant international stock exchange, hosting over 800 offshore listings with a combined market capitalization exceeding £ 140 billion (Gutierrez et al., 2024). These listings allow investors to diversify across local stocks registered in the UK and dual-class stocks listed offshore (Wahid et al., 2017).

However, AIM also introduces risks, such as reduced liquidity and lower disclosure requirements compared to traditional markets (Mortazian, 2022; Xue and O'Sullivan, 2023). Transactions in primary listing markets can influence trading patterns of cross-listed firms, adding complexity to portfolio efficiency.

AIM supports sustainable investing through its Green Economy Mark, awarded to firms generating over 50 % of revenue from environmentally friendly products or services. This initiative has recognized more than 100 issuers with a combined market capitalization of £ 149 billion, promoting investment in green IPOs and reinforcing AIM's role in advancing sustainable finance (Chouaibi

et al., 2022; London Stock Exchange, 2022).

3. Methodology

3.1. Sample and Data

This study evaluates the greenness of newly issued shares using firm-specific data from Reuters, spanning 2006–2020. Carbon emissions data prior to each firm's IPO date were averaged to assess environmental impact. The sample comprises 312 stocks, representing approximately 30 % of the 1051 IPOs issued on AIM during this period, as shown in Fig. 1. The starting year, 2006, reflects the earliest availability of emissions data, while the endpoint, 2020, ensures sufficient time for assessing long-term performance with a three-year post-IPO cutoff.

Service-providing industries, financial institutions, and IT sectors were excluded due to their categorization as low-carbon producers. These sectors contribute minimally to overall emissions compared to production-intensive industries (e.g., manufacturing, construction, and transport). Their inclusion could distort findings by introducing outliers with high sales volumes but low emissions. This exclusion aligns with studies demonstrating the negligible emissions from trade in services compared to goods (Ma and Wang, 2021) and illustrating the sectoral disparities in emissions intensity, where production-intensive industries are identified as major contributors to greenhouse gas emissions (Liu et al., 2012).

The final sample includes 139 cross-listed stocks: 21 from Australia, 16 from Bermuda, 20 from Canada, 29 from the Cayman Islands, 10 from Israel, 23 from Ireland, and 20 from the United States. The remaining 173 stocks are locally incorporated primary listings. Stock performance was evaluated over short-term (up to one month) and long-term (three years) holding periods to comprehensively analyze portfolio dynamics across different investment horizons.

3.2. Measuring greenness of stocks

This study introduces a Greenness Index to classify stocks based on their carbon emissions relative to production output. The index incorporates firm-level emissions of carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O), weighted by their global warming potential and normalized against sales volume to reflect production scale.

Previous studies have proposed metrics such as green indices (Mumtaz and Yoshino, 2021), green scores (Meriç et al., 2012; Prober et al., 2015), and pollution scores (Akhtaruzzaman et al., 2023) to classify stocks into green and non-green categories. Similarly, Rahat and Nguyen (2022) utilized carbon intensity and green revenue metrics to evaluate portfolio greenness. Building on these approaches, this study introduces a Greenness Index that incorporates firm-level emissions of CO₂, CH₄, and N₂O, normalized by sales volume to reflect production scale. By including sales volume, the Greenness Index provides a more comprehensive measure of emissions intensity, addressing limitations in prior methodologies reliant on absolute emissions.

The pollution score aggregates firm emissions:

$$\text{Pollution Score}_i = \sum_{g \in \{CO_2, CH_4, N_2O\}} (\text{Proportion of Emissions from Firm } i \times \text{Weight of } g) \quad (1)$$

This score is normalized into a Size-Based Pollution Score to account for production output:

$$\text{Size Based Pollution Score}_i = \frac{\text{Pollution Score}_i}{\text{Sale volume}_i} \quad (2)$$

The Greenness Index calculates a stock's emissions as a proportion of total emissions across all firms in the sample period:

$$\text{IPO's Portion}_{ij} = \frac{\text{Total emission produced by stock}_j}{\sum_{i=0}^{i=n} \text{Total emission produced by all listed stock}_i} \quad (3)$$

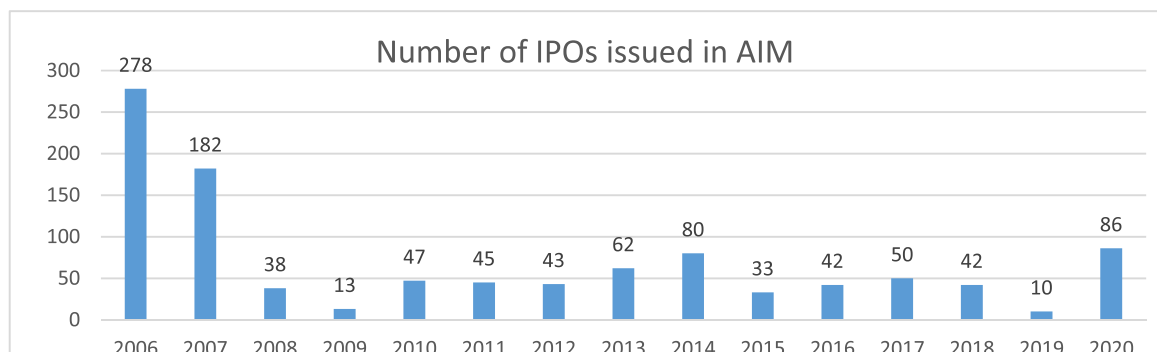


Fig. 1. Number of listed companies during 2006–2020 at AIM.

Using this methodology, stocks are classified as green (emissions below the median) or non-green (emissions at or above the median). Each category is further divided into High and Low subcategories by partitioning the sample into quartiles based on emissions magnitude.

By incorporating production scale, this methodology ensures a more accurate assessment of environmental impact compared to approaches relying solely on absolute emissions (Akhtaruzzaman et al., 2023). Applying the Greenness Index, the study identified 168 green stocks and 144 non-green stocks, providing a balanced framework for analysis.

3.3. Portfolio construction

Portfolios were constructed to analyze risk-return tradeoffs while incorporating the greenness classification. Following Markowitz's approach to diversification, the allocation strategy accounted for investor risk aversion and preferences for sustainable investments. Stocks were categorized into local and dual-class shares, further subdivided into green and non-green categories. The portfolio optimization process ensured alignment with the utility function:

$$U = R_t - \beta\sigma^2_t + \delta(\text{Domicile}_t) + \delta(\text{Green}_t) \quad (5)$$

where R_t represents the return of green, non-green, local, and dual-class stocks at time t . The term σ^2 reflects the risk associated with stocks, incorporating considerations of domicile and greenness. The parameter β indicates the investor's sensitivity to risk versus return, while δ captures the investor's preferences, emphasizing local and dual-class stocks within an eco-friendly framework.

Higher values of β signify greater risk aversion, as they reflect a stronger emphasis on minimizing risk over maximizing returns. In contrast, δ highlights the degree to which investors prioritize sustainability and domicile-related factors in their decision-making.

The technical details of portfolio optimization—including mathematical formulations for return and variance calculations, covariance matrices, and marginal effects of correlation on portfolio weights—are detailed in Appendix A.

3.4. Determinants of short-run returns

Numerous studies have examined the factors influencing short-run returns in various markets. For instance, Rock (1986) observed an investor tendency to pay higher prices for stocks in the secondary market, which causes higher initial returns due to information asymmetry (Pour, 2017). This phenomenon occurs because of investor knowledge disparities. Investors are categorized into informed and uninformed groups, with informed investors more likely to gather pertinent information (Sundarasan et al., 2018). Dodd and Gilbert (2016) suggest that informed investors estimate the intrinsic value of new issues based on available information. Alternatively, some theories propose that mispricing may occur due to insufficient knowledge about specific firm characteristics, such as Age, Size, and financial position (Abdul-Rahim et al., 2016).

The ex-ante uncertainty hypothesis, discussed in studies like Mantell (2016), posits that investor risk perception leads to mispricing and variable returns. This risk is categorized into pre- and post-IPO uncertainties, encompassing factors like Size, Offer Size, and Mkt Volat. Another perspective considers the impact of UW Prestige on short-run returns, suggesting that prestigious underwriters typically yield higher short-term gains (Arora and Singh, 2019; Khurshed et al., 2016).

Additionally, market factors also play a role in IPO underpricing, as demonstrated by the signaling hypothesis, where issuers may intentionally underprice to enhance their market stature (Badru and Ahmad-Zaluki, 2018; Colombo et al., 2019). The ownership dispersion hypothesis is used to explain the short-run returns of IPOs, aiming to broaden market participation among smaller shareholders, although the link between abnormal returns and ownership dispersion remains unproven. Furthermore, agency theory explores conflicts of interest between stakeholders and management (Hanafi and Setiawan, 2018; Pour, 2017), identifying a significant relationship between agency conflicts, insider holding, and short-run returns.

To identify the factors that influence short-run pricing performance, the following equation is developed to evaluate the hypotheses:

$$MAAR_{stock} = \alpha_0 + \partial_1 S_i + \partial_2 C_i + \partial_3 M_i + \partial_4 G_i + \beta_1 Z_i + \mu_i \quad (6)$$

$MAAR^1$ represents market-adjusted abnormal returns calculated over 30 days, assuming the investor holds the stock for this period. Here, S_i represents stock-specific characteristics, including risk (aftermarket risk, defined as the standard deviation of post-issue returns for 45 days), oversubscription (total issued shares divided by total demand), offer size (total shares issued), and waiting period (time between the issue date and listing on AIM). C_i includes firm-specific characteristics such as return on assets (net income divided by total assets), leverage (long-term debt divided by total assets), size (logarithm of total assets), and age (time elapsed from incorporation to the offering date). M_i represents market-specific characteristics, including market volatility and hot market conditions (a dummy variable equal to 1 if issued in a hot market and zero otherwise). G_i encompasses governance-related characteristics,

¹ Market Adjusted Abnormal Return = $\left[\left(\frac{1+R_{i,1}}{1+R_{m,1}} \right) - 1 \right] \times 100$ where $R_{i,1}$ is the raw return of stock at the end of first trading day. Total return is calculated as: $R_{i,1} = \frac{P_{i,1} - P_{i,0}}{P_{i,0}}$ where $P_{i,1}$ is the price of stock i at the close of the first trading day, $P_{i,0}$ is the offer price. $R_{m,1}$ is the market return measured during the corresponding period as: $R_{m,1} = \frac{I_{m,1} - I_{m,0}}{I_{m,0}}$ where $I_{m,1}$ is the market index at the close of first trading day and $I_{m,0}$ is the market index value on the offer day of the respective stock.

including board size (total number of board members), institutional shareholding (percentage of shares held by institutions), and CEO duality (an indicator for a CEO holding dual roles). Finally, Z_i captures offshore-related risk factors, including local currency returns, parental market returns, and exchange rate impacts from GBP and USD. This revised representation consolidates the variables into meaningful categories, ensuring clarity and brevity in the analysis.

3.5. Determinants of long-run returns

Ritter (1991) introduced the anomaly of long-run underperformance of new issues, linking it to the "window of opportunity" hypothesis. This theory suggests that during hot market periods, firms tend to overprice IPOs, resulting in lower long-run returns as intrinsic values correct over time. This is supported by studies such as Loughran and Ritter (1995), who observed that firms capitalize on favorable market conditions to raise capital at inflated prices, often leading to subpar performance later. Similarly, Zubair Mumtaz, Smith, and Ahmed (2016) reported that hot market issuances often deviate from investor expectations in the long run.

Another explanation is the 'impresario hypothesis,' which posits that investment banks underprice IPOs to generate demand, thereby enhancing their market reputation. Studies (Aggarwal et al., 1993; Aggarwal and Rivoli, 1990; Bondt and Thaler, 1985; Fama et al., 1969) show a strong correlation between initial underpricing and subsequent underperformance, as share prices adjust to intrinsic values.

The 'divergence of opinion hypothesis' provides further insights, highlighting that uncertainty surrounding IPOs often results in overvaluation during initial trading days. Miller (1977), (2001) argued that optimistic investor sentiment drives early demand, but as market realism sets in, prices correct, leading to underperformance. This divergence diminishes as differences in opinion converge, reflecting more accurate valuations (Kooli and Suret, 2004; Guo, Lev and Shi, 2006).

Agency and entrenchment theories also link managerial ownership to long-run performance. Increased managerial control can sometimes lead to negative valuation effects due to entrenched interests, particularly in family businesses (Mazzola and Marchisio, 2002; Sahoo and Rajib, 2010). Studies such as Chahine (2007) report significant underperformance for IPOs in family-run firms, influenced by these dynamics.

To evaluate these hypotheses, the following equation is developed:

$$BHAR_{stock} = \beta_0 + \beta_1 S_i + \beta_2 C_i + \beta_3 M_i + \beta_4 G_i + \beta_5 Z_i + \mu_i \quad (7)$$

Here, $BHAR^2$ represents the buy-and-hold abnormal returns over three years, assuming an investor holds the stock for the period. The vector S_i includes stock-specific characteristics such as risk, oversubscription, offer size, and waiting period. The vector C_i captures firm-specific characteristics, including ROA, leverage, firm age, and size. The vector M_i encompasses market-specific characteristics, such as market volatility and hot market conditions. The vector G_i reflects governance indicators, including board size, institutional shareholding, and CEO duality. Finally, the vector Z_i accounts for offshore risks, incorporating local currency returns, parental market returns, and exchange rate impacts from GBP and USD.

3.6. Econometric model

To enhance the precision of variable selection and ensure the robustness of the model, we employ LASSO (Least Absolute Shrinkage and Selection Operator) regression. This technique, introduced by Santosa and Symes (1986), is widely regarded for its ability to balance prediction accuracy and model interpretability, especially in settings with numerous predictors. The specification of the LASSO regression is as follows:

$$Return_i = \frac{1}{2m} \sum_{i=1}^m \left(y_i - \beta_0 + \sum_{j=1}^p x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (8)$$

In this formula, λ acts as a tuning parameter that controls the strength of the penalty applied to the model coefficients. When $\lambda = 0$, the regression behaves as a standard ordinary least squares (OLS) model, where no penalty is applied. Conversely, when $\lambda \rightarrow \infty$, all coefficients are shrunk to zero. For $0 < \lambda < \infty$, the model balances two objectives: minimizing residual sum of squares and applying a penalty to shrink coefficients, effectively selecting only the most relevant variables. The L_1 regularization penalty in LASSO causes some coefficients to shrink to zero, thereby achieving variable selection within the linear model. As λ increases, more coefficients are set to zero, resulting in fewer variables being included. Among the nonzero coefficients, greater shrinkage is applied as λ grows.

LASSO is preferred over OLS when the aim is to improve prediction accuracy and model interpretability, particularly in datasets with numerous predictors (Tibshirani, 1996). Unlike OLS, which may lead to overfitting in such contexts, LASSO imposes penalties on large coefficients, effectively enhancing robustness and focusing on the most influential variables.

² In this study, we use the Buy and Hold Abnormal Return (BHAR) technique to determine the long-run pricing performance of IPOs. Following Loughran and Ritter (1995), BHAR for firm i at time t is computed as: $BHAR_{it} = \prod_{t=1}^T [1 + R_{it}] - 1$ $BHAR_{it} = \frac{1}{n} \sum_{i=1}^n [\prod_{t=1}^T [1 + R_{it}] - \prod_{t=1}^T [1 + R_{mt}]]$

4. Results and discussion

4.1. Descriptive statistics

Table 1 provides the descriptive statistics for the sample of 312 stocks, including 173 locally incorporated stocks and 139 offshore or secondary listings on AIM. These stocks are categorized into green and non-green groups. Among the local stocks, 94 are green, with 43 classified as high green and 51 as low green, yielding average returns of 5.25 % and 1.26 %, respectively, over three years under a buy-and-hold strategy. Green stocks show lower short-term returns but significantly outperform non-green stocks in the long run.

In contrast, the 79 local non-green stocks consist of 30 low non-green stocks and 49 high non-green stocks, producing returns of −4.56 % and −0.75 %, respectively, over the same period. While non-green stocks provide better short-term returns, they underperform green stocks over longer periods.

For dual-class stocks, high green stocks yield the highest returns (7.01 %), followed by low green stocks (2.18 %), high non-green stocks (1.58 %), and low non-green stocks (−0.63 %). Dual-class stocks generally offer higher returns across all categories compared to local stocks, reflecting additional risks associated with foreign exchange and market dynamics in both home and host markets for secondary listings.

4.2. Portfolio optimization of local stocks in short-and long run

Table 2 reveals that in the short term, investors can achieve the highest return of 7.01 % by allocating 100 % of their portfolio to low non-green stocks. Conversely, the lowest return of −0.63 % occurs when fully investing in high green stocks. Assuming a buy-and-hold strategy for one month, an equally weighted portfolio delivers a 2.53 % return with a risk of 2.11 %. An optimal portfolio, providing a 5 % return with moderate risk, can be constructed by allocating 37.03 % of the portfolio to low green stocks and 62.97 % to low non-green stocks, as illustrated in Fig. 2.

The results indicate that green IPOs generally yield lower short-term returns due to the higher costs associated with renewable and sustainable processes. However, green stocks demonstrate a lower risk compared to their non-green counterparts, reflecting reduced underpricing driven by lower uncertainty and sustainability risks. Investors seeking to minimize short-term risk may find green portfolios, particularly within the local stock category, to be an attractive option (Mumtaz and Yoshino, 2021).

Table 3 demonstrates that in the long run, the portfolio with the lowest risk allocates 65.61 % to green stocks and 34.39 % to non-green stocks. For investors with higher risk tolerance, fully investing in low-green stocks offers the highest return of 5.25 % with a risk of 1.16 %, assuming the stocks are purchased on the first trading day and held for three years. By comparison, an equally weighted portfolio across the four categories—low green, high green, low non-green, and high non-green—yields a modest return of 0.30 % with a risk of 0.73 % over the long term. An optimal portfolio, delivering higher returns while maintaining moderate risk, is achieved by allocating 77.73 % to low green stocks, as depicted in Fig. 3.

The findings indicate that long-term portfolios heavily weighted toward green stocks provide higher expected returns while incurring lower risk. These results align with prior research by Anderloni and Tanda (2017) and Chan and Walter (2014), which suggest that while green technologies, products, and services may initially involve higher costs compared to non-green alternatives, they offer greater sustainability and eco-friendliness over time.

Table 1
Descriptive Statistics.

Domicile	Category	N	Carbon Emission	Returns	Offer Price	Offer Size	Market Capitalization
Local Stocks	Low Green	51	> 1146.788 < = 5436.75	5.25 % (10.61 %)	215.554 ** (140.183)	222.544 ** (329.875)	798.201 ** (1121.910)
	Low Non-Green	30	> 5436.75 < = 62811.335	−4.56 % (20.08 %)	215.504 ** (221.110)	321.611 ** (1211.915)	2149.533 ** (6849.097)
	High Green	43	= < 1146.788	1.26 % (6.65 %)	212.431 ** (311.231)	149.013 ** (130.712)	810.064 ** (781.670)
	High Non-Green	49	> 62811.335	−0.75 % (12.82 %)	210.331 ** (183.321)	226.815 ** (350.479)	788.771 ** (1132.791)
Dual Class IPO	Low Green	40	> 1146.788 < = 5436.75	2.18 % (25.59 %)	219.321 ** (136.231)	219.211 ** (311.125)	888.034 ** (1472.336)
	Low Non-Green	25	> 5436.75 < = 62811.335	−0.63 % (25.84 %)	221.611 ** (209.050)	362.656 ** (1153.405)	2421.211 ** (6711.217)
	High Green	34	= < 1146.788	7.01 % (39.55 %)	220.411 ** (324.211)	151.211 ** (101.902)	823.064 ** (899.312)
	High Non-Green	40	> 62811.335	1.58 % (15.63 %)	199.948 ** (176.577)	216.813 ** (332.221)	708.415 ** (1045.843)
	Total	312					

Source: AIM at LSE. (Monetary amounts of offer size and market capitalization are expressed in £,000,000)

Table 2
Short-Term Portfolio Returns and Marginal Effects of Local Green and Non-Green Stocks.

Stocks	Normal Returns	Lowest Returns	R= -0 %	R= 1 %	R= 2 %	R= 3 %	R= 4 %	R= 5 %	R= 6 %	Highest Returns
Portfolio Return	2.53 % (14.54 %)	−0.63 % (26.22 %)	0.00 % (19.85 %)	1.00 % (13.79 %)	2.00 % (14.27 %)	3.00 % (22.23 %)	4.00 % (21.52 %)	5.00 % (25.27 %)	6.00 % (32.44 %)	7.01 % (40.12 %)
Portfolio Risk	2.11 %	6.87 %	3.94 %	1.90 %	2.04 %	4.94 %	4.63 %	6.39 %	10.53 %	16.10 %
Risk Free Rate	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %
Sharp Ratio	16.74 %	−2.78 %	−0.50 %	6.53 %	13.31 %	13.05 %	18.12 %	19.39 %	18.18 %	17.23 %
Investment Allocation										
Low Green	25.00 %	0.00 %	13.76 %	29.35 %	48.04 %	0.00 %	0.00 %	37.03 %	18.62 %	0.00 %
Low Non-Green	25.00 %	0.00 %	0.00 %	3.24 %	4.03 %	17.05 %	37.73 %	62.97 %	81.38 %	100.00 %
High Green	25.00 %	100.00 %	74.67 %	41.28 %	2.99 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %
High Non-Green	25.00 %	0.00 %	11.57 %	26.13 %	44.94 %	82.95 %	62.27 %	0.00 %	0.00 %	0.00 %

The sample comprises 173 local stocks listed on AIM from 2006 to 2020. The portfolio is constructed based on the assumption that investors hold the stocks for a one-month period, representing short-run performance. The table presents the highest and lowest portfolio returns across different allocation categories, along with the marginal effects of adjustments in investor risk and return preferences.

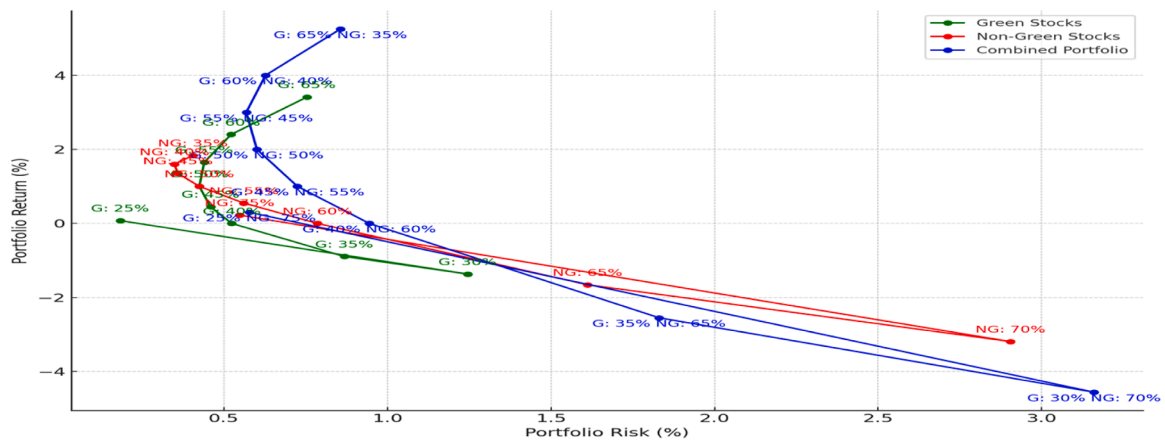


Fig. 2. Efficient Frontier Curve Based on Greenness in Local Stocks for the Short-Run.

Table 3

Long-Term Portfolio Returns and Marginal Effects of Local Green and Non-Green Stocks.

Stocks	Normal Returns	Lowest Returns	R= -2.55 %	R= 0 %	R= 1 %	R= 2 %	R= 3 %	R= 4 %	Highest Returns
Portfolio Return	0.30 % (8.54 %)	-4.56 % (20.38 %)	-2.55 % (15.74 %)	0.00 % (11.46 %)	1.00 % (10.11 %)	2.00 % (9.25 %)	3.00 % (8.96 %)	4.00 % (9.31 %)	5.25 % (10.76 %)
Portfolio Risk	0.73 %	4.15 %	2.48 %	1.31 %	1.02 %	0.85 %	0.80 %	0.87 %	1.16 %
Risk Free Rate	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %
Sharp Ratio	2.34 %	-22.85 %	-16.83 %	-0.87 %	8.90 %	20.55 %	32.38 %	41.90 %	47.84 %
Investment Allocation									
Low Green	25.00 %	0.00 %	13.61 %	11.28 %	27.74 %	44.35 %	61.00 %	77.73 %	100.00 %
Low Non-Green	25.00 %	100.00 %	73.03 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %
High Green	25.00 %	0.00 %	8.08 %	3.79 %	4.41 %	4.55 %	4.61 %	4.41 %	0.00 %
High Non-Green	25.00 %	0.00 %	5.28 %	84.93 %	67.85 %	51.10 %	34.39 %	17.86 %	0.00 %

The sample comprises 173 local stocks listed on AIM from 2006 to 2020. The portfolio is constructed with the assumption that the investor holds the stock for three years (long-run). The table highlights the highest and lowest portfolio returns with allocation to different categories and the marginal effects if investors adjust their risk and return preferences.

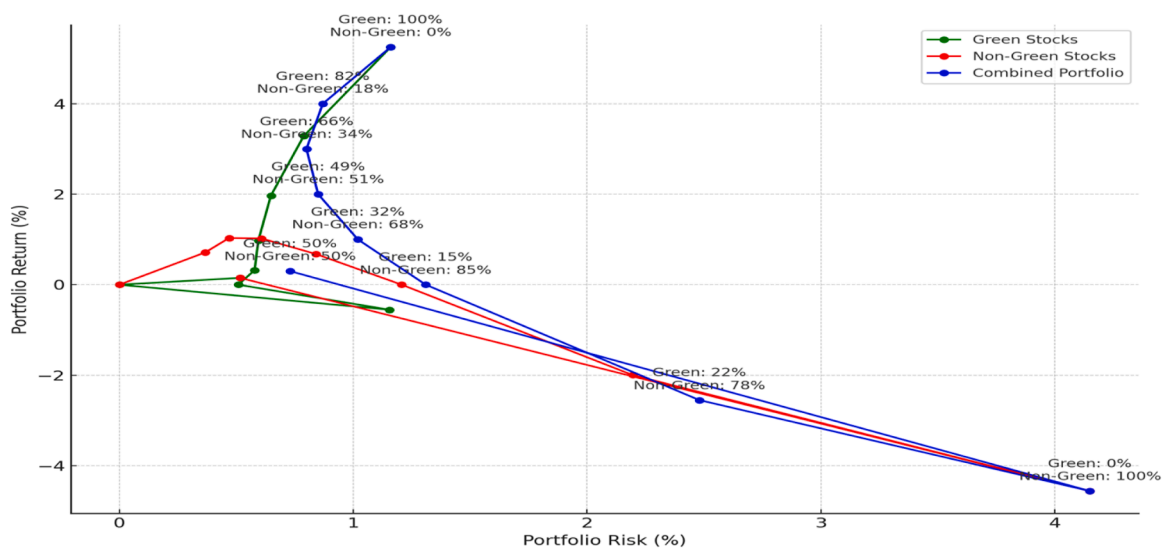


Fig. 3. Efficient Frontier Curve based on Greenness in local Stocks for long-run.

4.3. Portfolio optimization of dual class stocks in short-and long run

Table 4 shows that in the short run, an optimal portfolio achieves a 4 % return with 1.02 % risk by allocating 66 % of the portfolio to green stocks and 34 % to non-green dual-class stocks. For investors seeking the highest return, a 6.25 % return with 1.16 % risk is possible by fully investing in low non-green dual-class stocks. Conversely, the lowest return of 1.26 % with 0.46 % risk is observed when fully allocating to high green stocks. An equally weighted portfolio across green and non-green categories provides a 2.57 % return with 0.84 % risk, demonstrating that dual-class stocks generally yield higher returns than local stocks due to offshore risk factors.

Fig. 4 illustrates the efficient frontier, highlighting the trade-off between risk and return for dual-class stocks in the short run.

In the long run, as depicted in Table 5, the optimal portfolio achieves a 4 % return with a 6.37 % risk by allocating 78.44 % of funds to green stocks and 21.56 % to non-green stocks. The highest return of 6.01 %, with a risk of 16.10 %, is attained by fully investing in green dual-class stocks. Alternatively, the lowest return of −4.56 % is observed when the portfolio is fully allocated to high non-green stocks, accompanied by a 4.15 % risk. A balanced portfolio, equally weighted between green and non-green stocks, yields a 0.27 % return with 1.79 % risk, illustrating a conservative approach to long-term investment.

Fig. 5 presents the efficient frontier for long-run portfolio optimization, showing the optimal return-risk combinations based on investor preferences.

These findings suggest that green dual-class stocks, despite their relatively lower initial returns in the short term, provide sustainable and eco-friendly investment opportunities with higher long-term returns. The price variability of AIM-listed dual-class stocks offers diversification benefits, helping investors achieve superior long-term performance. However, challenges such as reduced liquidity, limited disclosure requirements, and the impact of primary market transactions on host market trading patterns should be considered when investing in AIM-listed dual-class stocks.

4.4. Descriptive analysis of determinants

Table 6 provides an overview of key variables characterizing Local and Dual-Class IPOs, with a focus on short-run (MAAR) and long-run (BHAR) performance. Overall, the data highlight significant differences in risk profiles, firm attributes, and market dynamics between the two categories.

Dual-Class IPOs exhibit higher *Risk* compared to Local IPOs, reflecting the greater uncertainty associated with their structure. Similarly, *Offer Size* and *Firm Size* are consistently larger for Dual-Class IPOs, suggesting that these firms tend to have greater capital-raising capacity and a more established market presence. Conversely, *Market Volatility* is higher for Local IPOs, indicating their greater sensitivity to market fluctuations.

Underwriter Prestige is more prominent among Local IPOs, signaling a preference for reputable underwriters in these offerings. However, *Hot Market* conditions are more favorable for Dual-Class IPOs, aligning with strategies to launch such offerings during optimal market environments. Governance differences are also evident, with *CEO Duality* being more common in Dual-Class IPOs, reflecting variations in corporate governance practices.

Additional variables, such as *Over Subscription*, *Leverage*, *Age*, *Board Size*, and *Institutional Shareholding*, exhibit distinct patterns between the two IPO types. For instance, Dual-Class IPOs demonstrate slightly higher demand relative to supply, as indicated by their *Over Subscription* rates, and are associated with larger and more established firms. In contrast, Local IPOs display greater variability in governance and market-related attributes.

These characteristics collectively underscore the contrasting dynamics shaping the performance and risk profiles of Local versus Dual-Class IPOs, providing insights into investor preferences and market behavior.

Table 4
Portfolio Returns and Marginal Effects of Dual-Class Stocks Splitting into Green and Non-Green Stocks in the Short Run.

Stocks	Normal Returns	Lowest Returns	R= 2 %	R= 3 %	R= 4 %	R= 5 %	Highest Returns
Portfolio Return	2.57 % (9.17 %)	1.26 % (6.75 %)	2.00 % (5.71 %)	3.00 % (11.64 %)	4.00 % (10.10 %)	5.00 % (10.40 %)	6.25 % (10.76 %)
Portfolio Risk	0.84 %	0.46 %	0.33 %	1.35 %	1.02 %	1.08 %	1.16 %
Risk Free Rate	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %
Sharp Ratio	26.89 %	17.22 %	33.27 %	24.92 %	38.63 %	47.12 %	47.84 %
Investment Allocation							
Low Green	25.00 %	0.00 %	17.49 %	38.74 %	66.00 %	93.25 %	0.00 %
Low Non-Green	25.00 %	100.00 %	77.10 %	0.00 %	0.00 %	0.00 %	100.00 %
High Green	25.00 %	0.00 %	4.01 %	0.00 %	0.00 %	0.00 %	0.00 %
High Non-Green	25.00 %	0.00 %	1.39 %	61.26 %	34.00 %	6.75 %	0.00 %

The study sample comprises 139 dual-class stocks listed on AIM from 2006 to 2020. The portfolio is constructed based on the assumption that investors hold the stocks for a one-year period, representing short-run performance. The table presents the highest and lowest portfolio returns, along with the marginal effects, illustrating the impact of adjustments in investor risk and return preferences on portfolio outcomes.

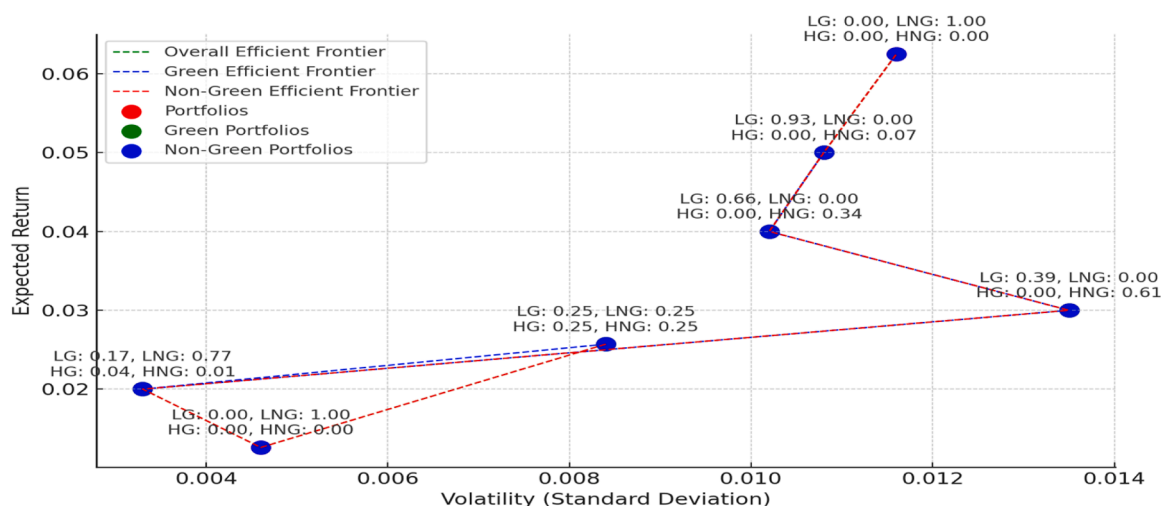


Fig. 4. Efficient Frontier Curve Based on Greenness in Dual-Class Stocks for the long Run.

4.5. Determinants of short-run returns of local and dual class stock

Table 7 highlights the determinants of short-run returns for high green, low green, low non-green, and high non-green stocks, along with the overall sample. Risk is significant at the 5 % level across all specifications, with positive coefficients, indicating that greater aftermarket risk is associated with higher short-run returns. This reflects a standard risk-return tradeoff, where investors require compensation for bearing higher levels of risk.

Oversubscription has a significant negative effect at the 5 % level in all categories, suggesting that highly oversubscribed IPOs tend to experience lower short-run returns. This finding aligns with the hypothesis that intense initial demand inflates prices, leading to subsequent corrections. Similarly, underwriter prestige is consistently significant at the 5 % level with positive coefficients, showing that IPOs managed by prestigious underwriters perform better in the short run due to reduced informational asymmetry and increased investor confidence.

Offer size is positively associated with short-run returns at the 5 % level across all categories, suggesting that larger IPOs attract more attention and resources, enhancing performance. Firm age is significant at the 5 % level for high green, low non-green, and the overall sample, indicating that older firms tend to achieve better short-run returns, likely due to their established reputation.

Firm size is significant at the 5 % level in all specifications, with positive coefficients, highlighting its role as a key determinant of performance. Larger firms are generally perceived as more stable and capable of delivering returns, which may explain their superior short-term outcomes. Conversely, market volatility negatively affects returns in several categories, with significant results for low green, low non-green, high non-green, and the overall sample, emphasizing the detrimental impact of market uncertainty on IPO performance.

Hot market conditions have a positive and significant effect on short-run returns at the 5 % level in all categories, supporting the "window of opportunity" hypothesis. This theory posits that firms take advantage of investor optimism during favorable market periods to achieve advantageous pricing.

Corporate governance factors also play a significant role. Board size is positively associated with short-run returns and is significant at the 5 % level for the overall sample, suggesting that stronger governance structures enhance performance. Institutional shareholding shows a positive relationship with returns at the 5 % level across most categories, indicating that higher ownership by institutional investors boosts confidence and financial discipline. In contrast, CEO duality has a negative and significant impact on returns at the 5 % level, underscoring concerns about governance conflicts when CEOs hold dual roles.

Table 8 explores the determinants of short-run returns for dual-class stocks, highlighting several important factors. Risk, oversubscription, and underwriter prestige are significant at the 5 % level across all specifications. Positive coefficients for risk and underwriter prestige indicate that higher risk and reputable underwriters contribute to stronger short-run performance, consistent with risk-return tradeoff theory and signaling effects. Oversubscription has a negative coefficient, reflecting the potential for price corrections following excessive demand during the IPO.

Firm age and firm size are significant at the 5 % level in all categories, demonstrating that older and larger firms tend to deliver higher short-run returns due to their perceived stability and operational strength. Financial leverage is positively significant for the overall sample, indicating its influence on investor perceptions of risk and return potential.

Hot market conditions positively influence short-run returns at the 5 % level, aligning with the "window of opportunity" hypothesis. Conversely, market volatility exhibits mixed effects, with limited significance across specifications, suggesting that its impact varies depending on stock characteristics.

Governance indicators such as board size, institutional shareholding, and CEO duality significantly affect returns. Larger boards are positively associated with performance, reflecting improved oversight and strategic decision-making. Institutional shareholding

Table 5

Portfolio Returns and Marginal Effects of Dual-Class Stocks Splitting into Green and Non-Green Stocks in the Long Run.

Stocks	Normal Returns	Lowest Returns	R= -2 %	R= 0 %	R= 1 %	R= 2 %	R= 3 %	R= 4 %	R= 5 %	Highest Returns
Portfolio Return	0.27 % (13.37 %)	−4.56 % (20.38 %)	−2.00 % (13.58 %)	0.00 % (14.93 %)	1.00 % (14.99 %)	2.00 % (16.31 %)	3.00 % (21.14 %)	4.00 % (25.24 %)	5.00 % (29.89 %)	6.01 % (40.12 %)
Portfolio Risk	1.79 %	4.15 %	1.84 %	2.23 %	2.25 %	2.66 %	4.47 %	6.37 %	8.93 %	16.10 %
Risk Free Rate	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %	0.10 %
Sharp Ratio	1.26 %	−22.85 %	−15.47 %	−0.67 %	6.00 %	11.65 %	13.72 %	15.45 %	16.40 %	17.23 %
Investment Allocation										
Low Green	25.00 %	0.00 %	3.72 %	8.88 %	21.90 %	34.98 %	47.86 %	60.87 %	73.87 %	100.00 %
Low Non-Green	25.00 %	0.00 %	27.27 %	50.87 %	42.23 %	29.73 %	29.88 %	21.56 %	13.80 %	0.00 %
High Green	25.00 %	0.00 %	27.72 %	40.25 %	35.87 %	35.29 %	22.27 %	17.57 %	12.33 %	0.00 %
High Non-Green	25.00 %	100.00 %	41.29 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %	0.00 %

The sample consists of 139 dual-class stocks listed on AIM from 2006 to 2020. The portfolio is constructed based on the assumption that investors hold the stocks for a three-year period, representing long-run performance. The table presents the highest and lowest portfolio returns, along with the marginal effects, illustrating how adjustments in investor risk and return preferences impact portfolio performance.

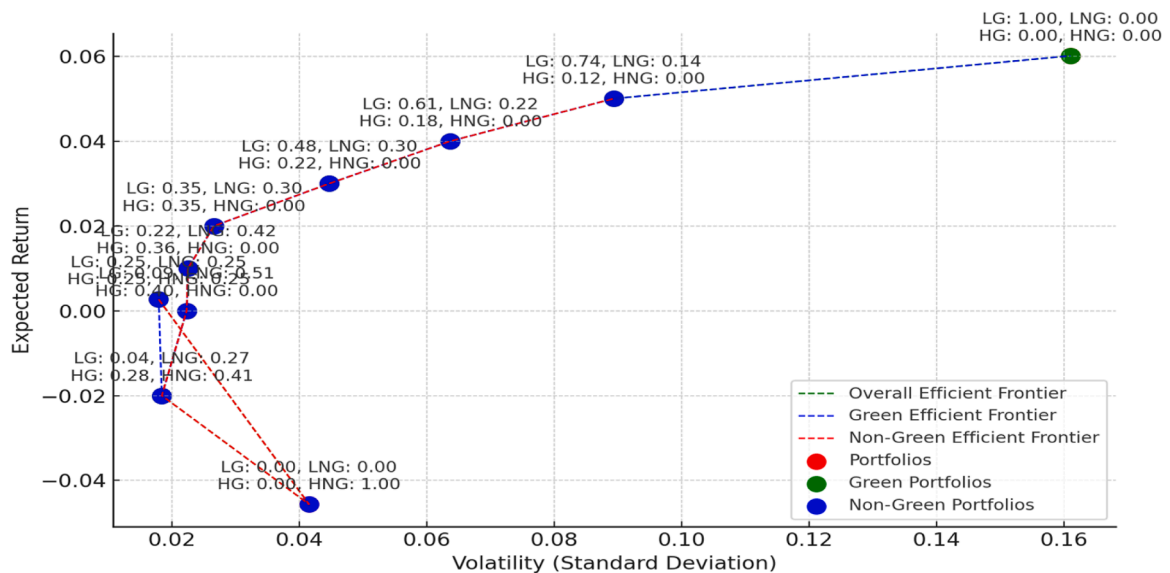


Fig. 5. Efficient Frontier Curve Based on Greenness in Dual-Class Stocks for the long run.

Table 6
Descriptive Analysis.

Characteristic	Local IPOs (MAAR)	Local IPOs (BHAR)	Dual-Class IPOs (MAAR)	Dual-Class IPOs (BHAR)
Risk	0.086 (0.033)	0.136 (0.045)	0.151 (0.047)	0.231 (0.053)
Over Sub.	-0.423 (0.176)	-0.321 (0.151)	-0.433 (0.110)	-0.110 (0.042)
UW Prestige	0.503 (NA)	0.503 (NA)	0.321 (NA)	0.321 (NA)
Offer Size	0.304 (0.314)	0.112 (0.121)	0.121 (0.349)	0.611 (0.210)
Wait Period	-0.204 (0.226)	-0.121 (0.041)	-0.321 (0.095)	-0.056 (0.019)
ROA	-0.342 (0.354)	NA	-0.123 (0.259)	NA
Leverage	0.694 (0.277)	-0.312 (0.091)	0.321 (0.072)	-0.121 (0.051)
Age	0.564 (0.196)	0.121 (0.064)	0.221 (0.090)	0.312 (0.073)
Size	0.211 (0.130)	0.312 (0.041)	0.812 (0.290)	0.039 (0.016)
Market Volatility	-0.609 (0.226)	-0.101 (0.021)	-0.121 (0.094)	-0.051 (0.027)
Hot Mkt	0.304 (NA)	-0.201 (NA)	0.812 (NA)	-0.612 (NA)
Board Size	0.086 (0.065)	0.109 (0.045)	0.167 (0.080)	0.401 (0.032)
Inst. Shareholding	0.423 (0.104)	0.112 (0.029)	0.321 (0.073)	0.281 (0.071)
CEO Duality	-0.292 (NA)	-0.121 (NA)	-0.711 (NA)	-0.001 (NA)

This table summarizes the characteristics of IPOs, including continuous variables such as Risk, Over Subscription, Offer Size, Waiting Period, ROA, Financial Leverage, Firm Age, Firm Size, Market Volatility, Board Size, and Institutional Shareholding. Additionally, binary variables such as Underwriter Prestige, Hot Market, and CEO Duality are also presented.

enhances performance, likely due to increased market confidence. CEO duality negatively affects returns, indicating potential governance inefficiencies when CEOs hold dual roles.

Offshore risk factors, including local currency returns, parental market returns, GBP returns, and USD returns, also significantly impact short-run returns. Parental market returns positively influence performance, suggesting that favorable home-market conditions boost investor confidence in host markets. In contrast, negative coefficients for GBP and USD returns indicate that currency fluctuations and related risks erode short-run returns for dual-class stocks.

The results underscore the complex interplay of factors influencing short-run returns on AIM. For both local and dual-class stocks, firm-specific attributes (age, size), market-specific conditions (hot markets), and governance indicators (board size, institutional shareholding) are key drivers of performance. However, dual-class stocks face additional offshore risks, emphasizing the importance of managing currency and market exposures in optimizing short-run returns.

4.6. Determinants of long-run returns of local and dual class stock

Table 9 indicates that aftermarket risk is a significant determinant of long-run returns for local stocks, with positive coefficients at the 5 % level across all specifications. This suggests that stocks with higher aftermarket risk yield greater returns in the long run, consistent with the risk-return tradeoff theory. Conversely, oversubscription shows a significant negative impact on long-run returns at the 5 % level. This finding implies that heightened demand during the IPO phase may result in price corrections over time, reducing

Table 7
Determinants of Short-run Returns of Local Stocks.

		High Green Stocks	Low Green Stocks	Low Non-Green Stocks	High Non-Green Stocks	Overall
Stocks Specific Characteristics	Risk	0.086 (5.97)* *	0.136 (3.96)* *	0.096 (3.68)* *	0.067 (3.55)* *	0.061 (3.88)* *
	Over sub.	-0.423 (2.99)* *	-0.483 (2.44)*	-0.642 (2.50)*	-0.161 (2.90)* *	-0.321 (3.61)* *
	UW Prestige	0.503 (3.12)* *	5.485 (3.61)* *	3.195 (4.23)* *	3.231 (4.02)* *	0.511 (3.32)* *
	Offer Size	0.304 (2.01)*	0.959 (2.19)*	0.663 (2.43)*	0.213 (2.91)* *	0.121 (2.21)*
	Wait Period	-0.204 (0.48)	-0.681 (0.38)	-0.623 (0.91)	-0.113 (1.57)	-0.121 (1.22)
	Firm's Specific Characteristics	ROA	-0.342 (1.41)	-0.615 (0.56)	-0.941 (0.49)	-0.567 (0.81)
Firm's Specific Characteristics	Leverage	0.694 (2.63)*	0.036 (1.07)	0.211 (1.55)	0.229 (0.55)	0.231 (1.23)
	Age	0.564 (2.94)* *	0.455 (1.53)	0.059 (2.09)*	0.032 (1.49)	0.488 (2.94)* *
	Size	0.211 (3.22)* *	0.551 (2.17)*	0.305 (2.85)* *	0.281 (2.98)* *	0.421 (3.22)* *
	Market Specific Characteristics	Mkt Volat.	-0.609 (0.86)	-0.288 (2.42)*	-0.848 (2.82)* *	-0.609 (1.86)
	Hot Mkt	0.304 (3.15)* *	0.119 (4.03)* *	0.389 (3.92)* *	0.321 (3.55)* *	0.311 (2.15)*
Corporate Governance Indicators	Board Size	0.086 (1.97)*	0.197 (0.12)	0.260 (0.21)	0.211 (1.21)	0.136 (3.97)* *
	Inst. Sh.	0.423 (2.99)* *	0.136 (3.96)* *	0.096 (3.68)* *	0.121 (4.68)* *	0.423 (3.31)* *
	CEO Duality	-0.292 (4.28)* *	-0.483 (4.99)* *	-0.642 (4.50)* *	-0.642 (5.50)* *	-0.421 (6.28)* *
	R ²	0.41	0.40	0.38	0.36	0.45
N		43	51	30	49	173

The sample consists of 173 local stocks listed on AIM from 2006 to 2020. Stock returns are calculated based on the assumption that an investor holds the stock for one month. The dependent variable, Monthly Average Abnormal Return (MAAR), is used in the analysis. All regressions include a constant term, which is not reported for brevity. Numbers in parentheses represent standard errors. * **, *, and * correspond to statistical significance at 1 %, 5 %, and 10 % levels, respectively.

long-term returns.

Initial underpricing is another significant negative factor at the 5 % level, reflecting its adverse influence on long-run performance. Stocks initially priced below their intrinsic value are more likely to experience a correction in returns over time, eroding long-term gains.

Firm-specific factors, including firm age and size, are positively associated with long-run returns at the 5 % level. Older firms benefit from established reputations, while larger firms are perceived as more stable, contributing to stronger long-term performance.

Hot market conditions exhibit a significant negative relationship with long-run returns at the 5 % level, supporting the "window of opportunity" hypothesis. Firms issuing stocks during market booms may achieve short-term gains but face subsequent corrections, diminishing long-term performance.

Corporate governance indicators, such as board size, institutional shareholding, and CEO duality, significantly affect long-run returns. Larger board sizes and higher institutional shareholding are positively associated with performance, indicating the importance of robust governance structures. In contrast, CEO duality negatively impacts returns at the 5 % level, highlighting governance conflicts and potential management entrenchment as risks to long-term performance.

Table 10 explores the determinants of long-run returns for dual-class stocks. Similar to local stocks, aftermarket risk emerges as a significant positive factor at the 5 % level, while oversubscription and initial underpricing negatively affect returns, also at the 5 % level. These results underscore the enduring influence of market dynamics during the IPO phase on long-term performance.

Firm age and size positively influence dual-class stock returns at the 5 % level. Older and larger firms tend to deliver stronger long-term returns, benefiting from investor confidence and operational resilience.

Market-specific characteristics, such as hot market conditions, are negatively associated with long-run returns at the 5 % level, reinforcing the risks of issuing stocks during favorable but potentially overheated market periods.

Corporate governance indicators play a critical role in the long-term performance of dual-class stocks. Board size and institutional shareholding positively affect returns at the 5 % level, suggesting that sound governance practices enhance financial stability and investor confidence. CEO duality, while generally negative, is not consistently significant, indicating that its impact on dual-class stocks may vary depending on other governance and market factors.

Offshore risk factors significantly influence dual-class stocks' long-run performance. Parental market returns have a significant positive impact at the 5 % level, reflecting the influence of home market dynamics on host market performance. In contrast, local currency, GBP, and USD returns exhibit significant negative coefficients, highlighting the risks associated with currency fluctuations

Table 8
Determinants of Short-run Returns of Dual Class Stocks.

		High Green Stocks	Low Green Stocks	Low Non-Green Stocks	High Non-Green Stocks	Overall
Stocks Specific Characteristics	Risk	0.151 (4.88)* *	0.122 (3.89)* *	0.113 (4.67)* *	0.167 (4.52)* *	0.061 (3.88)* *
	Over sub.	-0.433 (4.03)* *	-0.401 (3.90)* *	-0.422 (4.50)* *	-0.431 (3.90)* *	-0.431 (4.61)* *
	UW Prestige	0.321 (4.01)* *	0.121 (4.51)* *	0.195 (4.23)* *	0.121 (4.32)* *	0.012 (4.12)* *
	Offer Size	0.121 (1.01)	0.761 (1.65)	0.132 (4.13)* *	0.991 (3.91)* *	0.541 (3.51)* *
	Wait Period	-0.321 (1.19)	-0.211 (1.13)	-0.132 (1.22)	-0.231 (1.61)	-0.343 (2.17)*
Firm's Specific Characteristics	ROA	-0.123 (1.22)	-0.712 (1.23)	-0.212 (1.09)	-0.121 (1.08)	-0.211 (1.22)
	Leverage	0.321 (3.78)* *	0.211 (1.23)	0.310 (1.30)	0.391 (1.08)	0.331 (1.71)
	Age	0.221 (3.44)* *	0.231 (3.53)* *	0.131 (4.11)* *	0.201 (3.61)* *	0.291 (4.01)* *
	Size	0.812 (4.08)* *	0.311 (4.21)* *	0.167 (2.91)* *	0.178 (2.99)* *	0.919 (3.87)* *
Market Specific Characteristics	Mkt Volat	-0.121 (0.13)	-0.288 (1.42)	-0.321 (1.42)	-0.321 (1.82)	-0.314 (1.32)
	Hot Mkt	0.812 (4.54)* *	0.713 (4.89)* *	0.451 (5.11)* *	0.321 (3.89)* *	0.761 (4.23)* *
Corporate Governance Indicators	Board Size	0.167 (4.97)* *	0.231 (2.12)*	0.190 (1.21)	0.192 (1.81)	0.231 (4.97)* *
	Inst. Sh.	0.321 (2.80)* *	0.310 (3.09)* *	0.161 (3.89)* *	0.209 (4.08)* *	0.123 (3.43)* *
	CEO Duality	-0.711 (4.91)* *	-0.121 (4.12)* *	-0.321 (4.91)* *	-0.321 (6.09)* *	-0.311 (4.11)* *
Risk Factors associated with Offshore Listing	Loc. Curr.	-0.130 (23.73)* *	-0.111 (4.76)* *	-0.150 (4.75)* *	-0.036 (4.58)* *	-0.030 (5.19)* *
	Par. Mkt	0.370 (4.08)* *	0.315 (4.52)* *	0.372 (2.63)* *	0.319 (3.66)* *	0.382 (4.17)* *
	GBP	-0.058 (3.39)* *	-0.059 (43.87)* *	-0.058 (2.41)*	-0.026 (1.66)	-0.034 (7.33)* *
	USD	-0.178 (2.92)* *	-0.119 (1.08)	-0.134 (4.22)* *	-0.112 (2.41)*	-0.314 (3.41)* *
R ²		0.54	0.61	0.55	0.52	0.63
N		34	40	25	40	139

The sample consists of 139 dual class Stocks listed on AIM from 2006 to 2020. Stock returns are calculated based on the assumption that an investor holds the stock for one month. The dependent variable, Monthly Average Abnormal Return (MAAR), is used in the analysis. All regressions include a constant term, which is not reported for brevity. Numbers in parentheses represent robust standard errors. ***, **, and * correspond to statistical significance at 1 %, 5 %, and 10 % levels, respectively.

and international exposure.

The findings indicate that long-run returns for local and dual-class stocks are shaped by a combination of market-specific, firm-specific, and governance factors. For both stock categories, aftermarket risk, firm size, and board size positively influence performance, while oversubscription, initial underpricing, and hot market conditions negatively affect returns. Offshore risk factors are particularly relevant for dual-class stocks, emphasizing the complexities of international exposure in shaping long-term performance.

5. Conclusion

This study examines the performance of portfolios composed of local and dual-class stocks listed on AIM, with a focus on green stock allocation over short and long holding periods. In the short run, optimal portfolios tend to favor non-green stocks, which yield higher returns but at greater risk, driven by factors such as higher volatility and informational asymmetries. Conversely, green stocks outperform in the long run, offering better returns with reduced risk, emphasizing their role in sustainable investment strategies. Dual-class stocks generally deliver higher returns compared to local stocks, attributed to additional offshore risk exposure and broader market access.

Key determinants of performance include firm size and age, which are positively correlated with long-term returns, reflecting the stability and market confidence associated with established firms. In contrast, initial underpricing negatively impacts long-run performance, as prices correct over time. Green stocks on AIM exhibit lower risk levels but also lower short-run returns relative to non-green stocks, highlighting the trade-offs investors face when prioritizing sustainability.

AIM's supportive ecosystem, which caters to small enterprises and cross-listed stocks, contributes to enhanced long-term performance. The platform's initiatives, such as the Green Economy Mark, which certifies companies generating at least 50 % of their

Table 9
Determinants of Long-run Returns of Local Stocks.

		High Green Stocks	Low Green Stocks	Low Non-Green Stocks	High Non-Green Stocks	Overall
Stocks Specific Characteristics	Risk	0.136 (4.11)* *	0.134 (5.32)* *	0.135 (4.91)* *	0.138 (4.82)* *	0.136 (5.21)* *
	Over sub.	−0.321 (3.93)* *	−0.311 (4.11)* *	−0.321 (3.51)* *	−0.309 (5.11)* *	−0.350 (3.89)* *
	UW Prestige	−0.321 (5.01)* *	−0.121 (4.01)* *	−0.195 (5.11)* *	−0.121 (3.89)* *	−0.118 (4.09)* *
	Offer Size	0.112 (1.01)	0.131 (1.65)	0.190 (4.13)* *	0.091 (4.01)* *	0.154 (3.09)* *
	Wait Period	−0.121 (1.23)	−0.161 (1.23)	−0.191 (1.52)	−0.112 (1.11)	−0.134 (1.17)
Firm's Specific Characteristics	ROA	−0.312 (3.78)* *	−0.121 (1.14)	−0.291 (1.50)	−0.312 (1.71)	−0.121 (1.71)
	Leverage	0.121 (4.12)* *	0.101 (4.11)* *	0.101 (3.78)* *	0.119 (3.89)* *	0.116 (4.33)* *
	Age	0.312 (5.11)* *	0.311 (4.09)* *	0.321 (4.01)* *	0.309 (4.07)* *	0.309 (4.67)* *
Market Specific Characteristics	Size	−0.101 (0.13)	−0.121 (1.11)	−0.106 (1.12)	−0.112 (1.21)	−0.110 (1.09)
	Mkt Volat	−0.201 (3.54)* *	−0.209 (4.11)* *	−0.251 (4.34)* *	−0.291 (4.11)* *	−0.219 (4.61)* *
Corporate Governance Indicators	Hot Mkt	0.109 (4.61)* *	0.231 (4.01)* *	0.201 (5.01)* *	0.213 (5.81)* *	0.291 (5.07)* *
	Board Size	0.112 (3.61)* *	0.110 (3.19)* *	0.121 (3.12)* *	0.109 (4.23)* *	0.113 (5.11)* *
	Inst. Sh.	−0.121 (4.31)* *	−0.201 (5.09)* *	−0.301 (4.31)* *	−0.390 (5.11)* *	−0.291 (3.33)* *
R ²		0.54	0.53	0.49	0.46	0.71
N		34	40	25	40	139

The sample comprises 173 local stocks listed on AIM from 2006 to 2020. Stock returns are calculated based on the assumption that an investor holds the stock for a period of three years, representing the long-run performance. The dependent variable is the three-year Buy-and-Hold Abnormal Return (BHAR) for each category. All regressions include a constant term, which is not reported for brevity. Numbers in parentheses represent robust standard errors. * **, * *, and * correspond to statistical significance at 1 %, 5 %, and 10 % levels, respectively.

revenue from environmentally sustainable activities, underscore its commitment to promoting green investment.

From an investor perspective, the findings underline the importance of balancing risk and return objectives while integrating environmental considerations into portfolio strategies. The superior long-run performance of green stocks reinforces their appeal for sustainability-focused investors, even as short-term returns may lag their non-green counterparts.

Future research could investigate transitions between non-green and green stocks, delve deeper into service industry stocks, or conduct comparative sectoral analyses. Such studies could enhance understanding of the evolving dynamics of sustainable investment across various industries and geographies.

Author contributions

Adul Wahid was responsible for the conceptualization of the manuscript, the development of the methodology, and the data collection. Oskar Kowalewski contributed by analyzing the findings, revising, and editing the document. All authors have reviewed and approved the final version of the manuscript for submission.

Informed consent statement

Not applicable.

Funding

This study was made possible by the financial support of the National Science Centre, Poland, research grant OPUS 17 No. 2019/33/B/HS4/00369, which was awarded to Oskar Kowalewski. Abdul Wahid's involvement as a Postdoctoral Fellow was instrumental in the successful execution of the research.

CRediT authorship contribution statement

Wahid Abdul: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Kowalewski Oskar:** Writing – review & editing.

Table 10
Determinants of Long-run Returns of Dual Class Stocks.

		High Green Stocks	Low Green Stocks	Low Non-Green Stocks	High Non-Green Stocks	Overall
Stocks Specific Characteristics	Risk	0.231 (3.22)* *	0.241 (4.41)* *	0.271 (5.33)* *	0.281 (5.11)* *	0.221 (5.61)* *
	Over sub.	-0.110 (5.13)* *	-0.101 (3.31)* *	-0.109 (4.81)* *	-0.110 (3.44)* *	-0.115 (5.56)* *
	UW Prestige	-0.312 (4.31)* *	-0.334 (5.34)* *	-0.411 (3.44)* *	-0.310 (5.45)* *	-0.341 (4.22)* *
	Offer Size	0.611 (5.01)* *	0.681 (4.65)* *	0.701 (5.13)* *	0.591 (5.22)* *	0.612 (5.31)* *
	Wait Period	-0.056 (1.09)	-0.065 (1.01)	-0.067 (1.34)	-0.077 (1.09)	-0.091 (1.76)
	ROA	-0.121 (1.78)	-0.212 (1.34)	-0.201 (1.81)	-0.211 (1.63)	-0.110 (1.09)
	Leverage	0.312 (3.92)* *	0.391 (3.34)* *	0.362 (4.09)* *	0.332 (4.72)* *	0.321 (5.32)* *
Firm's Specific Characteristics	Age	0.039 (4.23)* *	0.029 (3.19)* *	0.035 (4.33)* *	0.034 (4.12)* *	0.039 (4.30)* *
	Size	-0.051 (0.99)	-0.067 (1.54)	-0.056 (1.01)	-0.053 (1.44)	-0.091 (1.08)
Market Specific Characteristics	Mkt Volat	-0.612 (4.04)* *	-0.601 (4.01)* *	-0.622 (4.90)* *	-0.581 (4.23)* *	-0.590 (4.32)* *
	Hot Mkt	0.401 (5.01)* *	0.409 (4.60)* *	0.420 (4.34)* *	0.406 (3.09)* *	0.441 (4.66)* *
Corporate Governance Indicators	Board Size	0.281 (7.01)* *	0.277 (6.91)* *	0.216 (6.81)* *	0.201 (6.75)* *	0.209 (8.09)* *
	Inst. Sh.	-0.001 (0.31)	-0.009 (0.19)	-0.007 (0.67)	-0.101 (1.11)	-0.091 (1.33)
	Loc. Curr.	-0.671 (23.73)* *	-0.891 (4.76)* *	-0.701 (4.75)* *	-0.903 (4.58)* *	-0.921 (5.55)* *
Risk Factors associated with Offshore Listing	Par. Mkt	0.511 (8.90)* *	0.509 (7.90)* *	0.499 (9.78)* *	0.498 (8.66)* *	0.671 (9.88)* *
	GBP	-0.121 (13.21)* *	-0.290 (12.07)* *	-0.201 (8.09)* *	-0.251 (8.09)* *	-0.209 (8.11)* *
	USD	-0.812 (12.02)* *	-0.831 (14.18)	-0.780 (14.22)* *	-0.931 (12.81)* *	-0.910 (13.41)* *
	R ²	0.69	0.61	0.66	0.63	0.62
N		34	40	25	40	139

The sample comprises 139 dual class stocks listed on AIM from 2006–2020. Stock returns are calculated based on the assumption that an investor holds the stock for a period of three years, representing the long-run performance. The dependent variable is the three-year Buy-and-Hold Abnormal Return (BHAR) for each category. All regressions include a constant term, which is not reported for brevity. Numbers in parentheses represent robust standard errors. * *, *, and * correspond to statistical significance at 1 %, 5 %, and 10 % levels, respectively.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

Appendix A. Technical Framework for Portfolio Optimization

Portfolio construction aims to mitigate risk and enhance investment sustainability, addressing two key challenges: selecting stocks from a diverse range of options and optimally allocating investments across categories. Previous studies highlight that risk-averse investors benefit from diversification strategies that protect assets from contemporary risks.

Steinert and Crowe (2001) emphasize the benefits of diversification and recommend using Markowitz's approach for portfolio optimization. This framework is further supported by Oliva and Renò (2018), who underscore its relevance for enhancing portfolio efficiency. Diversification strategies have also been shown to reduce portfolio risk as the number of securities increases (Kisaka et al., 2015). However, Fletcher (2017) notes that stock selection strategies can sometimes fail to provide significant benefits compared to random subsets, highlighting the complexity of optimization.

Several studies explore the influence of external factors on portfolio construction. Lee et al. (2019) and Mensi et al. (2015) examine how parental market dynamics, host market conditions, and global factors affect the risk-return tradeoff. Al Janabi (2022) demonstrates the diversification benefits of mixed commodity-stock portfolios using machine learning, supporting effective risk management strategies. Cesari and Cremonini (2003) advocate for rebalancing strategies that optimize portfolios under different market conditions, even when accounting for transaction costs.

Given the complexities of risk and return tradeoffs, we model the utility function as:

$$U = R_t - \beta\sigma_t^2 + \delta(\text{Domicile}_t) + \delta(\text{Green}_t) \quad (\text{A1})$$

where R_t represents the return of green, non-green, local, and dual-class stocks at time t . The term σ^2 reflects the risk associated with stocks, incorporating considerations of domicile and greenness. The parameter β indicates the investor's sensitivity to risk versus return, while δ captures the investor's preferences, emphasizing local and dual-class stocks within an eco-friendly framework.

Higher values of β signify greater risk aversion, as they reflect a stronger emphasis on minimizing risk over maximizing returns. In contrast, δ highlights the degree to which investors prioritize sustainability and domicile-related factors in their decision-making.

For a portfolio with n stock categories, the allocation is defined as:

$$\sum_{i=1}^n \lambda_i = 1 \quad (\text{A2})$$

Where λ_i is the proportion of the portfolio allocated to stock category i .

The return of the portfolio (r_p) is calculated as:

$$r_p = \lambda_{LHG}r_{LHG} + \lambda_{LLG}r_{LLG} + \lambda_{LHNG}r_{LHNG} + \lambda_{LLNG}r_{LLNG} + \lambda_{DHG}r_{DHG} + \lambda_{DLG}r_{DLG} + \lambda_{DHNG}r_{DHNG} + \lambda_{DLNG}r_{DLNG} = \sum_{i=1}^n \lambda_i R_i \quad (\text{A3})$$

where $\lambda_{LHG}, \lambda_{LLG}, \lambda_{LHNG}, \lambda_{LLNG}, \lambda_{DHG}, \lambda_{DLG}, \lambda_{DHNG}, \lambda_{DLNG}$ represent the proportions of portfolio assets allocated to the stock categories: Local High Green (LHG), Local Low Green (LLG), Local High Non-Green (LHNG), Local Low Non-Green (LLNG), Dual-Class High Green (DHG), Dual-Class Low Green (DLG), Dual-Class High Non-Green (DHNG), and Dual-Class Low Non-Green (DLNG). The returns from these categories are denoted by $r_{LHG}, r_{LLG}, r_{LHNG}, r_{LLNG}, r_{DHG}, r_{DLG}, r_{DHNG}, r_{DLNG}$, corresponding to the respective performance of each stock group in the portfolio.

The portfolio variance σ_p^2 is calculated as:

$$\sigma_p^2 = \text{Var}[R_p] = \sum_{i=1}^n \sum_{j=1}^n \lambda_i \lambda_j \sigma_{ij} \quad (\text{A4})$$

Where σ_{ij} represents the covariance between stock categories i and j , and $\sigma_{ii} = \sigma_i^2$ is the variance of category i . The standard deviation σ_p is calculated as $\sqrt{\sigma_p^2}$.

The matrix representation of portfolio returns and covariances enables comprehensive evaluation of diversification strategies across the eight stock categories. This structured approach highlights the interdependence between asset returns, facilitating informed decision-making for portfolio optimization.

	1	2	3	4	5	6	7	8
1	$\lambda_{LHG}r_{LHG}$	$\lambda_{LHG}r_{LLG}$	$\lambda_{LHG}r_{LHNG}$	$\lambda_{LHG}r_{LLNG}$	$\lambda_{LHG}r_{DHG}$	$\lambda_{LHG}r_{DLG}$	$\lambda_{LHG}r_{DHNG}$	$\lambda_{LHG}r_{DLNG}$
2	$\lambda_{LLG}r_{LHG}$	$\lambda_{LLG}r_{LLG}$	$\lambda_{LLG}r_{LHNG}$	$\lambda_{LLG}r_{LLNG}$	$\lambda_{LLG}r_{DHG}$	$\lambda_{LLG}r_{DLG}$	$\lambda_{LLG}r_{DHNG}$	$\lambda_{LLG}r_{DLNG}$
3	$\lambda_{LHNG}r_{LHG}$	$\lambda_{LHNG}r_{LLG}$	$\lambda_{LHNG}r_{LHNG}$	$\lambda_{LHNG}r_{LLNG}$	$\lambda_{LHNG}r_{DHG}$	$\lambda_{LHNG}r_{DLG}$	$\lambda_{LHNG}r_{DHNG}$	$\lambda_{LHNG}r_{DLNG}$
4	$\lambda_{LLNG}r_{LHG}$	$\lambda_{LLNG}r_{LLG}$	$\lambda_{LLNG}r_{LHNG}$	$\lambda_{LLNG}r_{LLNG}$	$\lambda_{LLNG}r_{DHG}$	$\lambda_{LLNG}r_{DLG}$	$\lambda_{LLNG}r_{DHNG}$	$\lambda_{LLNG}r_{DLNG}$
5	$\lambda_{DHG}r_{LHG}$	$\lambda_{DHG}r_{LLG}$	$\lambda_{DHG}r_{LHNG}$	$\lambda_{DHG}r_{LLNG}$	$\lambda_{DHG}r_{DHG}$	$\lambda_{DHG}r_{DLG}$	$\lambda_{DHG}r_{DHNG}$	$\lambda_{DHG}r_{DLNG}$
6	$\lambda_{DLG}r_{LHG}$	$\lambda_{DLG}r_{LLG}$	$\lambda_{DLG}r_{LHNG}$	$\lambda_{DLG}r_{LLNG}$	$\lambda_{DLG}r_{DHG}$	$\lambda_{DLG}r_{DLG}$	$\lambda_{DLG}r_{DHNG}$	$\lambda_{DLG}r_{DLNG}$
7	$\lambda_{DHNG}r_{LHG}$	$\lambda_{DHNG}r_{LLG}$	$\lambda_{DHNG}r_{LHNG}$	$\lambda_{DHNG}r_{LLNG}$	$\lambda_{DHNG}r_{DHG}$	$\lambda_{DHNG}r_{DLG}$	$\lambda_{DHNG}r_{DHNG}$	$\lambda_{DHNG}r_{DLNG}$
8	$\lambda_{DLNG}r_{LHG}$	$\lambda_{DLNG}r_{LLG}$	$\lambda_{DLNG}r_{LHNG}$	$\lambda_{DLNG}r_{LLNG}$	$\lambda_{DLNG}r_{DHG}$	$\lambda_{DLNG}r_{DLG}$	$\lambda_{DLNG}r_{DHNG}$	$\lambda_{DLNG}r_{DLNG}$

Optimizing Stock Portfolio Weights

A critical aspect of portfolio construction is determining the proportion of each stock category within the overall portfolio. This allocation depends on the investor's utility function, where the sum of portfolio weights equals one. The combination of stocks in a portfolio reflects the investor's risk aversion, which can involve two or more stock categories (Bhuyan et al., 2014). In this study, we focus on two primary categories: local and dual-class stocks, further subdivided into green and non-green stocks.

To calculate the weight of local stocks in the portfolio, we use the following formula:

$$\lambda_{\text{Local stock}} = \frac{[r_L - r_D + \gamma(\sigma_L^2 - \sigma_{LD})] - [\lambda_L * \gamma(\sigma_L^2 - \sigma_{LD})]}{\gamma(\sigma_L^2 + \sigma_D - 2\sigma_{LD})} \quad (\text{A5})$$

If the investor chooses not to invest in local stocks, the proportion allocated to dual-class stocks is computed as:

$$\lambda_{\text{Dual Class stock}} = \frac{[r_D - r_L + \gamma(\sigma_D^2 - \sigma_{DL})] - [\lambda_L * \gamma(\sigma_D^2 - \sigma_{DL})]}{\gamma(\sigma_D^2 + \sigma_L - 2\sigma_{DL})} \quad (\text{A6})$$

With the weights of individual stock categories known, the overall portfolio weight is determined as:

$$\lambda_{\text{Portfolio}} = \frac{[r_L - r_D + \gamma(\sigma_L^2 - \sigma_{LD})] \rho - [r_D - r_L + \gamma(\sigma_D^2 - \sigma_{DL})] \beta}{\gamma(\rho - \beta^2)} \quad (\text{A7})$$

Here,

$$\rho = (\sigma_L^2 + \sigma_D^2 - 2\sigma_{LD}) \text{ and } \beta = (\sigma_D^2 - \sigma_{DL}).$$

The same methodology applies when further dividing local and dual-class stocks based on greenness.

Marginal Effects of Stock on Portfolio

To explore the maximum and minimum levels of risk and return, we construct various portfolio configurations and examine how these affect the proportion of each stock category in the overall portfolio. The relationship between the weights of local stocks (λ_L) and dual-class stocks (λ_{LD}) is given by:

$$\frac{\lambda_L}{\lambda_{LD}} = \frac{1}{\gamma(\sigma_L^2 + \sigma_D^2 - 2\rho_{DL}\sigma_L\sigma_D)} > 0 \quad (\text{A8})$$

The marginal effects of changes in the correlation (ρ_{DL}) on the optimal portfolio weights for local and dual-class stocks, further categorized into green and non-green stocks, are expressed as:

$$\frac{\lambda_D}{\lambda_{DL}} = \frac{\sigma_D\sigma_L}{\sigma_D^2 + \sigma_L^2 - 2\rho_{DL}\sigma_L\sigma_D} \left[\frac{2[r_L - r_D + \gamma(\sigma_L^2 - \rho_{DL}\sigma_L\sigma_D)]}{\gamma(\sigma_L^2 + \sigma_D^2 - 2\rho_{DL}\sigma_L\sigma_D)} \right] \quad (\text{A9})$$

Eq. (A9) highlights the impact of changes in the correlation between categories (ρ_{DL}) on the optimal portfolio configuration. This marginal effect is determined by the variance and covariance of stock returns, individual risk preferences, and investor constraints.

Data availability

The authors do not have permission to share data.

References

- Abdul-Rahim, R., Che Yahya, N., & Yong, O. (2016). Evidence of Winner's Curse in Malaysian IPOs: 2000-2013. Available at SSRN 2719423.
- Aggarwal, R., Leal, R., Hernandez, L., 1993. The aftermarket performance of initial public offerings in Latin America. *Lahore J. Econ.* 22 (1), 42–53.
- Aggarwal, R., Rivoli, P., 1990. Fads in the initial public offering market? *Financ. Manag.* 19 (4), 45–57.
- Akhtaruzzaman, M., Banerjee, A.K., Boubaker, S., Moussa, F., 2023. Does green improve portfolio optimisation? *Energy Econ.* 124, 106831.
- Akhtaruzzaman, M., Banerjee, A.K., Ghaddallou, W., Umar, Z., 2022. Is greenness an optimal hedge for sectoral stock indices? *Econ. Model.* 117, 106030.
- Al Janabi, M.A., 2022. Optimization algorithms and investment portfolio analytics with machine learning techniques under time-varying liquidity constraints. *J. Model. Manag.* 17 (3), 864–895.
- Aliu, F., Nuhui, A., Krasniqi, B.A., Jusufi, G., 2021. Modeling the optimal diversification opportunities: the case of crypto portfolios and equity portfolios. *Stud. Econ. Financ.* 38 (1), 50–66.
- Anderloni, L., Tanda, A., 2017. Green energy companies: stock performance and IPO returns. *Res. Int. Bus. Financ.* 39, 546–552.
- Antony, A., 2020. Behavioral finance and portfolio management: Review of theory and literature. *J. Public Aff.* 20 (2), e1996.
- Arora, N., Singh, B., 2019. Impact of auditor and underwriter reputation on underpricing of SME IPOs in India. *Manag. Labour Stud.* 44 (2), 193–208.
- Awiszus, K., Capponi, A., Weber, S., 2022. Market efficient portfolios in a systemic economy. *Oper. Res.* 70 (2), 715–728.
- Badru, B.O., Ahmad-Zaluki, N.A., 2018. Explaining IPO initial returns in Malaysia: ex ante uncertainty vs signalling. *Asian Rev. Account.* 26 (1), 84–106.
- Beloskar, V.D., Nageswara Rao, S.V.D., 2024. Screening activity matters: Evidence from ESG portfolio performance in an emerging market. *Int. J. Financ. Econ.* 29 (3), 2593–2619.
- Bhuyan, R., Kuhle, J., Ikromov, N., Chiemeke, C., 2014. Optimal portfolio allocation among REITs, stocks, and long-term bonds: an empirical analysis of U.S. financial markets. *J. Math. Financ.* 4 (2), 104–112.
- Bondt, W.F.M.De, Thaler, R., 1985. Does the stock market overreact? *J. Financ.* 40 (3), 793.
- Cesari, R., Cremonini, D., 2003. Benchmarking, portfolio insurance, and technical analysis: A Monte Carlo comparison of dynamic strategies of asset allocation. *J. Econ. Dyn. Control* 27 (6), 987–1011.
- Chahine, S., 2007. Block-holder ownership, family control, and post-listing performance of French IPOs. *Manag. Financ.* 33 (6), 388–400.
- Chan, P.T., Walter, T., 2014. Investment performance of “environmentally-friendly” firms and their initial public offers and seasoned equity offers. *J. Bank. Financ.* 44 (1), 177–188.
- Chaweewanchon, A., Chaysiri, R., 2022. Markowitz mean-variance portfolio optimization with predictive stock selection using machine learning. *Int. J. Financ. Stud.* 10 (3), 64.
- Chouaibi, S., Chouaibi, J., Rossi, M., 2022. ESG and corporate financial performance: The mediating role of green innovation—UK common law versus Germany civil law. *Eur. J. Bus.* 17 (1), 46–71.
- Cipiloglu Yildiz, Z., Yildiz, S.B., 2022. A portfolio construction framework using LSTM-based stock markets forecasting. *Int. J. Financ. Econ.* 27 (2), 2356–2366.
- Colombo, M.G., Meoli, M., Vismara, S., 2019. Signaling in science-based IPOs: The combined effect of affiliation with prestigious universities, underwriters, and venture capitalists. *J. Bus. Ventur.* 34 (1), 141–177.
- Dai, Z., Kang, J., 2022. Some new efficient mean-variance portfolio selection models. *Int. J. Financ. Econ.* 27 (4), 4784–4796.
- Demiralay, S., Bayraci, S., 2021. Should stock investors include cryptocurrencies in their portfolios after all? Evidence from a conditional diversification benefits measure. *Int. J. Financ. Econ.* 26 (4), 6188–6204.
- Dobrynskaya, V. (2015). Upside and Downside Risks in Momentum Returns. Higher School of Economics Research Paper No. WP BRP, 50.
- Dodd, O., Gilbert, A., 2016. The impact of cross-listing on the home market's information environment and stock price efficiency. *Financ. Rev.* 51 (3), 299–328.
- Du, M., Zhang, X., 2020. Urban greening: A new paradox of economic or social sustainability? *Land Use Policy* 92, 104487.
- Fama, E.F., Fisher, L., Jensen, M.C., Roll, R., 1969. The adjustment of stock prices to new information. *Int. Econ. Rev.* 10 (1), 1–21.

- Fisher, J.D., D'Alessandro, J., 2021. Portfolio upside and downside risk—both matter. *J. Portf. Manag.* 47 (10), 158–171.
- Fjesme, S.L., 2020. Foreign market portfolio concentration and performance. *Financ. Manag.* 49 (1), 161–177.
- Fletcher, J., 2017. Exploring the benefits of using stock characteristics in optimal portfolio strategies. *Eur. J. Financ.* 23 (3), 192–210.
- Gunjan, A., Bhattacharyya, S., 2023. A brief review of portfolio optimization techniques. *Artif. Intell. Rev.* 56 (5), 3847–3886.
- Guo, R.J., Lev, B., Shi, C., 2006. Explaining the Short-and Long-Term IPO Anomalies in the US by R&D. *J. Bus. Financ. Account.* 33 (3-4), 550–579.
- Gutierrez, E.F., Minutti-Meza, M., Tatum, K., & Vulcheva, M. (2024). Consequences of expanded audit reports: Evidence from the United Kingdom's alternative investment market. Available at SSRN 3805879.
- Hanafi, M.M., Setiawan, A., 2018. Ownership concentration, institutional ownership, and IPO underpricing: evidence from Indonesia Stock Exchange. *Int. J. Gov. Financ. Inter.* 1 (1), 3–17.
- Hoque, A., Rakhi, S., Hassan, K., Le, T., 2020. The performance of stock portfolios: Evidence from analysing Malaysia case, and implication for open innovation. *J. Open Innov.: Technol., Mark., Complex.* 6 (4), 1–13.
- Horn, M., Oehler, A., 2024. Constructing stock portfolios by sorting on ESG ratings: Does the rating provider matter? *Int. Rev. Financ. Anal.* 96, 103568.
- Huang, H.H., Chang, T.H., Wang, C.P., 2024. Investment performance comparison among various portfolio selection strategies in Taiwan stock market. *Asia Pac. Manag. Rev.* 29 (2), 195–214.
- Jawad, M., Naz, M., Muqaddus, H., 2024. A multi-criteria decision-making approach for portfolio selection by using an automatic spherical fuzzy AHP algorithm. *J. Oper. Res. Soc.* 75 (1), 85–98.
- Khurshed, A., Kostas, D., Saadouni, B., 2016. Warrants in underwritten IPOs: the Alternative Investment Market (AIM) experience. *J. Corp. Financ.* 40, 97–109.
- Kisaka, S.E., Aloise, J., Hilary, M., 2015. Determining the optimal portfolio size on the Nairobi Securities Exchange. *J. Financ. Econ.* 6 (6), 215–230.
- Kooli, M., Suret, J.M., 2004. The aftermarket performance of initial public offerings in Canada. *J. multinatl. financ. manag.* 14 (1), 47–66.
- Lee, P.L., Chin, L., Law, S.H., Azman-Saini, W.N.W., 2019. Analysing the effect of portfolio concentration index and stock market correlation. *J. Ekon. Malays.* 53, 2.
- Liu, Z., Geng, Y., Lindner, S., Guan, D., 2012. Uncovering China's greenhouse gas emission from regional and sectoral perspectives. *Energy* 45 (1), 1059–1068.
- London Stock Exchange. (2022). Green Economy Mark: London leads the world's sustainable finance ecosystem. London Stock Exchange.**
- Loughran, T., Ritter, J.R., 1995. The new issues puzzle. *J. Financ.* 50 (1), 23–51.
- Ma, T., Wang, Y., 2021. Globalization and environment: Effects of international trade on emission intensity reduction of pollutants causing global and local concerns. *J. Environ. Manag.* 297, 113249.
- Mantell, E.H., 2016. A theory of underwriters' risk management in a firm-commitment initial public offering. *Rev. Quant. Financ. Account.* 46 (1), 179–193.
- Mazzola, P., Marchisio, G., 2002. The role of going public in family businesses' long-lasting growth: a study of Italian IPOs. *Fam. Bus. Rev.* 15 (2), 133–148.
- Mensi, W., Hammoudeh, S., Kang, S.H., 2015. Precious metals, cereal, oil and stock market linkages and portfolio risk management: Evidence from Saudi Arabia. *Econ. Model.* 51, 340–358.
- Meric, I., Watson, C.D., Meric, G., 2012. Company green score and stock price. *Int. Res. J. Financ. Econ.* 82, 15–23.
- Miller, E.M., 1977. Risk, uncertainty, and divergence of opinion. *J. Financ.* 32 (4), 1151–1168.
- Miller, E.M., 2001. Why the low returns to beta and other forms of risk. *J. Portf. Manag.* 27 (2), 40.
- Mortazian, M., 2022. Liquidity and volatility of stocks moved from the Main Market to the Alternative Investment Market (AIM). *Asia-Pac. Financ. Mark.* 29 (2), 195–220.
- Mumtaz, M.Z., Yoshino, N., 2021. Greenness index: IPO performance and portfolio allocation. *Res. Int. Bus. Financ.* 57, 101398.
- Oliva, I., Renò, R., 2018. Optimal portfolio allocation with volatility and co-jump risk that Markowitz would like. *J. Econ. Dyn. Control* 94, 242–256.
- Pour, E.K., 2017. Entering the public bond market during the financial crisis: Underinvestment and asymmetric information costs. *Res. Int. Bus. Financ.* 39, 102–114.
- Prober, L., Meric, I., Meric, G., 2015. Does company green score affect stock price? *Rev. Bus. Financ. Stud.* 6 (3), 11–19.
- Rahat, B., Nguyen, P., 2022. Risk-adjusted investment performance of green and black portfolios and impact of toxic divestments in emerging markets. *Energy Econ.* 116, 106423.
- Rahiminezhad Galankashi, M., Mokhtab Rafiei, F., Ghezelbash, M., 2020. Portfolio selection: a fuzzy-ANP approach. *Financ. Innov.* 6 (1), 17.
- Ritter, J.R., 1991. The long-run performance of initial public offerings. *J. Financ.* 46 (1), 3–27.
- Rock, K., 1986. Why new issues are underpriced. *J. Financ. Econ.* 15 (1–2), 187–212.
- Sahoo, S., Rajib, P., 2010. Aftermarket pricing performance of initial public offerings (IPOs): Indian IPO market 2002–2006. *Vikalpa* 35 (4), 27–43.
- Santosa, F., Symes, W.W., 1986. Linear inversion of band-limited reflection seismograms. *SIAM J. Sci. Stat. Comput.* 7 (4), 1307–1330.
- Steinert, M., Crowe, S., 2001. Global real estate investment: Characteristics, optimal portfolio allocation and future trends. *Pac. Rim Prop. Res. J.* 7 (4), 223–239.
- Sundarasan, S.D., Khan, A., Rajangam, N., 2018. Signaling roles of prestigious auditors and underwriters in an emerging IPO market. *Glob. Bus. Rev.* 19 (1), 69–84.
- Ta, V.-D., Liu, C.-M., Tadesse, D.A., 2020. Portfolio optimization-based stock prediction using long-short term memory network in quantitative trading. *Appl. Sci.* 10 (2), 437.
- Thakkar, A., Chaudhari, K., 2021. A comprehensive survey on portfolio optimization, stock price and trend prediction using particle swarm optimization. *Arch. Comput. Methods Eng.* 28 (4), 2133–2164.
- Tibshirani, R., 1996. Regression shrinkage and selection via the lasso. *J. R. Stat. Soc.* 58 (1), 267–288.
- Wahid, A., Talib, N., Naqvi, S.A.A., 2017. Strategic Change in Operating Trends of Public Listed Companies and Its Impact on Stock Market Growth. *Pak. Dev. Rev.* 369–383.
- Xue, B., O'Sullivan, N., 2023. The determinants of audit fees in the alternative investment market (AIM) in the UK: Evidence on the impact of risk, corporate governance and auditor size. *J. Int. Account., Audit. Tax.* 50, 100523.
- Zaimovic, A., Omanovic, A., Arnaut-Berilo, A., 2021. How many stocks are sufficient for equity portfolio diversification? A review of the literature. *J. Risk Financ. Manag.* 14 (11), 551.
- Zhou, X., Zhang, R., Polochova, V., 2021. The dependence of Chinese green and conventional bond markets under extreme and normal market conditions. *Environ. Sci. Pollut. Res.* 28 (38).
- Zubair Mumtaz, M., Smith, Z.A., Ahmed, A.M., 2016. The aftermarket performance of initial public offerings in Pakistan. *Lahore J. Econ.* 21 (1), 23–68.